

A certified abstention band for card-fraud decisions, and two quantum arms reported as measured

Quantum Daemons — Global Quantum + AI Challenge 2026, HSBC track

Code and every table behind this document:

github.com/thedaemon-wizard/Global_Quantum_AI_Challenge_HSBC

1 The decision, and what is missing from it

An issuer authorising a card transaction has three actions, not two: approve, decline, or challenge the cardholder through 3-D Secure. The middle one is what makes the problem tractable at scale — routing even one percent of authorisation volume to human analysts is not operationally possible, while a step-up challenge is. Fraud models are reported by AUC or average precision; neither is a statement about a decision, and neither carries a finite-sample guarantee. Conformal methods reach this problem closely — Singh et al. pair class-conditional prediction with cost-controlled abstention, Zhu et al. certify a false-negative rate on card fraud, Nayak and Bushara report a fraud objective their own paper calls conformal-style rather than a finite-sample guarantee, and Chen et al. carry the temporal case to a graph model — but none conditions the guarantee on the abstention region it routes to, and those that do order time certify a prediction set rather than a decision. Selection-conditional risk control does condition on the retained region (Xu et al.; Bai and Jin), under exchangeability and not on payments. Both are what this submission adds, and the second turns out to matter.

What we contribute. A three-valued rule whose *abstention band* carries a distribution-free, finite-sample risk certificate under a pre-registration frozen before any model was fitted, and an honest measurement of where two quantum approaches sit against a strong classical baseline on the same decision.

What a successful PoC would demonstrate. A step-up decision whose false-decline rate carries a distribution-free, finite-sample certificate at a level the bank sets, validated on a block it never saw — the artefact a model-risk function can audit, which an AUC cannot provide. This study meets that target on public data, at loose levels only ($\alpha \geq 0.10$) and at the band budgets the sample size permits; the PoC target is the same property at the budget the business chooses rather than the one the data forces.

What the rule does to a day of traffic. On the held-out block, at the certified configuration with the tightest α : 86.72% approved outright, 9.35% routed to a 3-D Secure challenge, 3.93% declined. The middle number is the whole argument for a three-valued rule — routing that volume to human analysts is not operationally possible and a step-up is — and it is the number a bank can price before it believes anything else in this document.

Expected impact. The deliverable is an operating guarantee, not a score, and three consequences follow from that — none of which needs a quantum result. A model-risk function can review a bound instead of a metric, which is what validation and monitoring SR 26-2 asks of a decisioning model and an AUC cannot supply — and it matters more here, not less, because fraud detection reads on the EU AI Act Annex III *exception*, so no external high-risk regime supplies that discipline. The operating point becomes an input: α and the band budget are chosen by the business, and the certificate reports whether that choice is attainable rather than asserting it. And the band bounds what the expensive component costs — Section 5 prices it against the authorisation budget — which is what makes a quantum re-ranker discussable at all. The statement prices the other side itself — \$443bn in 2021 falsely declined globally, 40%+ of customers abandoning after one — and a certified bound on the rate is what turns that into a number a bank can act on. We attach no figure of our own: the issuer-side cost is not public, so the study certifies the *rate* and leaves the bank to multiply it.

Neither quantum arm produced a usable result, and they failed differently. A fidelity quantum kernel was rejected by its own a-priori screens before it was run on the task — a decisive negative. A matrix-product-state classifier ran and did not beat gradient boosting, but that comparison is underpowered, so what it yields is a non-superiority bound rather than a null result.

What we do not claim. No quantum advantage of any kind. No portfolio-level PSD2 compliance figure — the SCA-RTS ceilings govern exemption eligibility, not a decline threshold, and the appendix records that category error as one we made and withdrew. And no claim that the certificate binds the *unconditional* false-decline rate: it does not, and Section 2 says why that is the point rather than a caveat.

2 Method

Paradigm. Hybrid classical-quantum: a *gate-based* fidelity kernel and a *quantum-inspired* matrix-product-state classifier, both confined to an abstention band and wrapped in a classical certificate that depends on neither.

The rule, and which block may touch which parameter. Figure 1 is the whole method. A classical scorer f splits traffic three ways against thresholds fixed in advance; the band is $B = \{x : \tau_{lo} \leq f(x) < \tau_{hi}\}$ and g is the in-band re-scorer, where a quantum model would enter if one were usable. None was, so the certified rule takes $g = f$ and λ is a third threshold on the same score. Which block sets what carries the guarantee: edges and thresholds from $\mathcal{D}_{band}$, λ certified on $\mathcal{D}_{cal}$, nothing selected on $\mathcal{D}_{test}$.

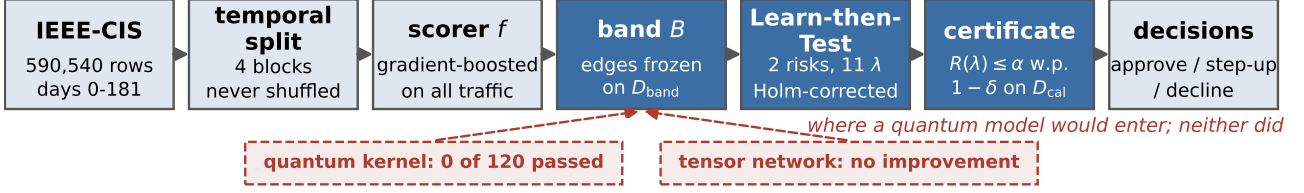


Figure 1: The method end to end; shaded stages carry the guarantee. Counts come from `splits.csv`. Neither quantum arm reached the decision.

The estimand is band-conditional, deliberately. We certify

$$R(\lambda) = \mathbb{P}(D(X) = \text{DECLINE} \mid Y = 0, X \in B),$$

not the marginal false-decline rate. The marginal rate is dominated by τ_{hi} and would be almost unchanged if g were replaced by a coin flip. A certificate that does not bind the component it is said to license is not evidence about that component — so the harder, smaller-sample quantity is the one we bound, and the cost of that choice is reported in Section 3 rather than hidden.

Two distinct breaks of exchangeability, and the split separates them. The blocks are 60/10/10/20 %, contiguous and snapped to day boundaries. The **temporal** break is that $\mathcal{D}_{cal}$ precedes $\mathcal{D}_{test}$. The **selective** break is that band membership is a data-dependent event, if the edges were estimated on the data used to certify. Freezing the edges on $\mathcal{D}_{band}$ removes the second by construction: membership becomes a fixed predicate, and filtering by one preserves exchangeability of what remains. Only the temporal break survives, and we measure it.

Risk control. Learn-then-Test controls family-wise error over a finite grid fixed in advance, using Hoeffding–Bentkus p -values with Holm. Two risks are controlled jointly: $R(\lambda)$ at level α , and the false-negative rate at α_{FN} . A grid point is admissible only when both nulls are rejected, so the joint statement inherits one confidence: $\mathbb{P}(R(\lambda) \leq \alpha) \geq 1 - \delta$ at $\delta = 0.05$.

The bound applies only to a *mean of per-observation losses*; our first formulation was a nonlinear transform of one and produced invalid p -values. It is now a 0/1 loss averaged over frauds, a recall floor at $1 - \alpha_{FN}$.

Coverage is judged against the exact law, not against α . For a sound split-conformal system the test-error count follows $E \sim \text{BetaBinomial}(m, n+1-k, k)$ with $k = \lceil (n+1)(1 - \alpha) \rceil$. Checking $\hat{r} \leq \alpha$ is a one-sided test against the wrong null: on a correctly calibrated system it passes only about half the time. Every verdict below is the two-sided Beta-Binomial interval at the 99 % level.

Imbalance is absorbed by thresholds, not by resampling. At a 3.412 % positive rate that is a choice, and the guarantee makes it: resampling the calibration block would break the exchangeability the certificate needs, so every block is scored as it falls and the imbalance is carried by where τ_{lo} , λ and τ_{hi} sit.

Intervals resample entities, not rows. The IEEE-CIS label propagates a reported chargeback to later transactions sharing an account, email or billing address, so rows are clustered: 85.0 % of test-block `card1` values also appear in training. Every interval is a card-level block bootstrap.

3 Results

What a random split buys you, measured. Table 1 runs one gradient-boosted model at fixed hyperparameters over five seeds on three splits of the same file. A stratified random split reports 0.3164 more average precision than a forward holdout — 0.8254 against 0.5090, non-overlapping across seeds. That is the largest effect in this study, larger than anything either quantum arm could have contributed, and it is why what follows reports a temporal split.

Table 1: The same gradient-boosted model, the same hyperparameters, five seeds, evaluated on the test block of three different splits. One model on one file, so it establishes the size of the effect here rather than a general rule, and it is the control arm of the split design rather than an extra experiment.

Split	ROC AUC	AP	AP vs forward holdout
stratified	0.9658	0.8254	+0.3164
temporal	0.8851	0.5090	—
card-disjoint	0.8509	0.5394	+0.0304

Table 2: Coverage across five seeds per arm and level, judged against the exact Beta-Binomial interval. It replaces a single-split table that reported an across-seed maximum as a centre; the appendix records that retraction. The temporal arm breaches on every seed at the two loosest levels; the stratified arm breaches on none at any level.

Arm	α	Ratio (mean)	sd	Seeds outside
temporal	0.010	1.436	0.055	5 of 5
temporal	0.005	1.456	0.036	5 of 5
temporal	0.002	1.314	0.067	4 of 5
temporal	0.001	0.943	0.020	0 of 5
stratified	0.010	1.008	0.038	0 of 5
stratified	0.001	0.962	0.140	0 of 5
card-disjoint	0.010	0.916	0.133	2 of 5
card-disjoint	0.001	0.882	0.362	1 of 5

It is also what separates this baseline from the published leaderboard. The 1st-place IEEE-CIS solution reports ROC AUC 0.9459 on the private leaderboard — the competition’s own held-out file — and about 0.9363 in time-based cross-validation over the training file with their UID feature removed, which they report as worth about +0.011 AUC — against 0.8837–0.8884 here on a strict forward holdout of that training file. Different tasks, measured above rather than assumed.

Coverage fails under time, on every seed, and the controls do not reproduce it. Table 2 calibrates on $\mathcal{D}_{\text{cal}}$ and reports on $\mathcal{D}_{\text{test}}$, five seeds per arm and level against the exact interval. The temporal arm falls outside on *every* seed at the two loosest levels and on 4 of 5 at the third, inside at the tightest; the stratified arm falls outside on none, at any level.

The card-disjoint arm is not clean either way: it *over*-covers on 2 of 5 seeds at $\alpha = 0.01$ and *under*-covers on 1 at $\alpha = 0.001$, with a spread (sd 0.362) an order of magnitude wider than the temporal arm’s, so it is not reported as a pass. Neither control reproduces the temporal arm’s every-seed breach, which is weaker than “the breach is temporal” because the arms differ in more than time. Label censoring is not detected separately: the trailing-window rate is 3.666 % against 3.281 % earlier.

But the breach is origin-dependent, and this narrows the claim. Re-running at five rolling calibration origins gives Table 3: the ratio ranges from 0.684 to 1.411. 2 of 5 breach, and the earliest is *conservative* — the deviation reverses direction. So the defensible statement is not a fixed inflation factor but that the deviation depends on where the calibration window falls, which one origin cannot establish.

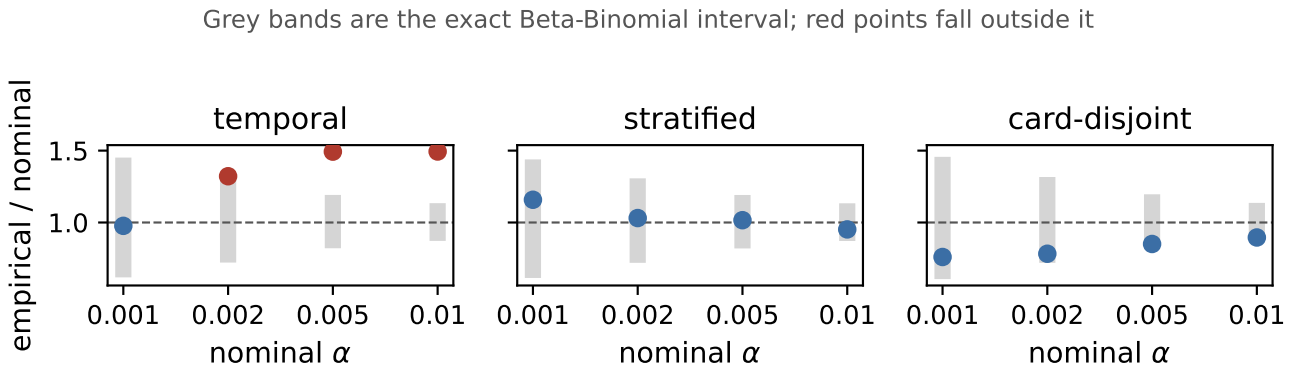


Figure 2: One seed, with the exact Beta-Binomial interval to scale; Table 2 gives all five. The interval widens sharply as α tightens, which is why the temporal arm’s compliance at the tightest level is weak evidence.

Table 3: The temporal deviation across five rolling calibration origins. 2 of 5 breach and the earliest is conservative, so no single ratio characterises the effect.

Origin (day)	Ratio	Errors	Verdict (interval)
101	0.684	379	conservative ([472, 641])
111	0.893	459	inside ([437, 595])
121	1.411	777	breached ([468, 638])
131	1.198	696	breached ([494, 674])
141	1.126	607	inside ([457, 625])

And it holds on the held-out block. This is H5. All 5 certified configurations were applied to $\mathcal{D}_{\text{test}}$ unchanged — same edges, same λ — and 5 of 5 realised a band-conditional false-decline rate at or below their certified α , the tightest using 0.856 of its budget. Holding a ceiling that loose is a real check that the machinery transfers and a weak one, which amendment A1 predicted before the arm ran.

What the certificate actually reaches. 5 of 48 configurations certify, all strictly inside the band — the smallest margin below τ_{hi} is 0.0136. That is not automatic: a λ at the band edge flags nothing, so its risk is structurally zero and the guarantee vacuous — as an earlier version of this table was.

The survivors need $\alpha \geq 0.10$ and $\alpha_{\text{FN}} = 0.45$, and nothing certifies at the tightest band budget — where $\mathcal{D}_{\text{cal}}$ contributes only 847 legitimate in-band rows against 58,343 overall. The binding mechanism is sample size acting through the concentration bound, not the class-conditional degeneracy floor, which is what makes it addressable: Yu and Liu report an empirical-Bernstein bound on selected risk beating Hoeffding in scoped regimes.

4 Two quantum arms, and how each failed

The kernel never reached the task. Before running a fidelity quantum kernel we pre-registered screens over 120 configurations of encoding (z, zz and dense-angle feature maps) $\times$ qubits $\times$ bandwidth $\times$ entanglement (linear or none), and gated on two: a kernel is usable only if it is not exponentially concentrated — effective rank $r_{\text{eff}} = \exp(-\sum_i p_i \log p_i)/n$ at $p_i = \sigma_i / \sum_j \sigma_j$, inside a usable band — *and* not reproducible by a tuned RBF ($\rho_{\text{RBF}} \leq 0.60$, the largest absolute off-diagonal correlation over a bandwidth grid).

28 of 120 pass conditioning. **0 pass both.** The closest any configuration came to distinctness was $\rho_{\text{RBF}} = 0.6291$; the median is 0.9740 and the maximum 0.99999975. On 300 stratified rows with at most eight features, no configuration cleared the pre-registered bar — which reporting a downstream accuracy would have hidden. One caveat: the protocol names the Huang geometric difference first and the implementation computes it without gating on it (amendment A6), which would have made this rejection less decisive, not more. The statement cites two positive results, 0.87 precision and 0.88 F_1 (Deloitte and AWS, 2024; El Alami et al., 2024, “Karimi” there). The second is a 984-row balanced undersample at seven principal components with no classical arm — neither this task’s 3.412 % base rate nor a comparison — and the first is a thresholded precision on a 0.172 % base rate — 3.412 % here — with no recall. Neither tests distinctness from a classical kernel before scoring, which is where this section stops. The one controlled benchmark on real card-transaction data (Faryad, arXiv:2608.15718, 2026) finds no robust quantum advantage, and one apparent advantage to be an artefact of unequal search budgets.

The tensor network ran, and did not win. A matrix-product-state classifier contracts

$$f_{\ell}(\mathbf{x}) = \sum_{\{s\}} [A^{s_1} \cdots A^{s_N}]_{\alpha} W_{\alpha\ell} \prod_j \phi^{s_j}(x_j), \quad \phi(x) = [\cos \frac{\pi x}{2}, \sin \frac{\pi x}{2}],$$

where W is the head that closes the final bond into the two class logits ℓ , sweeping the bond dimension χ rather than tuning it. In the band, on identical rows and features, it reaches average precision 0.1235 at the largest bond dimension against 0.1407 for the gradient-boosted baseline, a paired difference of -0.0172 . Both sit near the 10.96 % floor average precision cannot fall below, and neither ranks the band far above chance — the band is the region the incumbent scorer already could not separate, so re-scoring it has limited headroom for *any* model.

At full scale it loses again, with replication behind it. On all 431 features, trained on $\mathcal{D}_{\text{train}}$ and scored once on $\mathcal{D}_{\text{test}}$, across 16 fits — four bond dimensions at four seeds each — the best draw reaches 0.2512 average precision against 0.5055–0.5114 for the baseline — 49.1 % of it. The baseline also sees entity-aggregate columns the arm does not; without them the gap barely moves (docs/RESULTS.md). The in-band 87.8 % is not comparable with it: average precision floors at the base rate, and the band is 10.96 % positive against 3.412 %. As lift over that floor the two agree, 44.8 % and 45.5 %, so the deficit does not widen with dimension. Every one of the 16 loses, so which draw is reported does not change the conclusion.

Table 4: H4 in the band: matrix-product-state against the gradient-boosted baseline on identical rows and features. Intervals are 95 % card-clustered bootstraps. Every interval contains zero, and the comparison is underpowered against its pre-registered ceiling, so what is established is a non-superiority bound rather than a null.

χ	AP (MPS)	AP (GBDT)	Δ AP	95 % CI	p
4	0.1202	0.1407	-0.0205	$[-0.0593, +0.0149]$	0.864
8	0.1173	0.1407	-0.0234	$[-0.0630, +0.0122]$	0.895
16	0.1183	0.1407	-0.0225	$[-0.0613, +0.0136]$	0.884
32	0.1235	0.1407	-0.0172	$[-0.0582, +0.0200]$	0.806

Capacity is not resolvable, and replication is how we know. The largest seed spread at one bond dimension is 0.1781 average precision against 0.0629 for the capacity signal — 2.83 times — so a per- χ ordering is selection on a maximum. **And 2 of 16 fits diverged to the prior**, at one seed and two bond dimensions, on the stable contraction path; a matched re-run reproduces both to six decimals. Excluding them *raises* the ratio to 3.76, so part of the apparent capacity signal was the failures.

The comparison is underpowered, and the gate that should have said so was itself wrong. It is a comparison between model *families*, whose measured standard error is 0.0199 — 8.03-fold the baseline-against-itself figure the gate originally used. The effect it can resolve is 0.0557 average precision, against a pre-registered ceiling of 0.023; the appendix gives both defects in full. So H4 is reported as underpowered, which is what the protocol commits to. Every interval contains zero, so any improvement is bounded above by +0.0200 average precision — a non-superiority bound, not evidence of equivalence and not evidence that the tensor network is worse.

5 Feasibility, resources, and the constraints we already know

What the PoC sprint would build. The same protocol on the issuer’s stream: fit the incumbent scorer, freeze four blocks, calibrate the band, certify the band-conditional false-decline rate at a level the business picks. Every component has been run end to end here, so the sprint is a port and a re-calibration rather than a build: three stages of roughly a month with a month of slack — split integrity, then calibration and certification, then the quantum re-ranker, last because it is the only stage that can fail to produce a result.

Compute, measured rather than estimated. The classical core is cheap: the baseline fits the in-band problem in under a second, and calibration is a sort. The expensive component is the quantum-inspired arm: 16 full-scale tensor-network fits over 356,216 rows and 431 sites took 2685–2783 seconds each on the one RTX PRO 6000 Blackwell (96 GB), CUDA 13, Python 3.12 workstation every number here was measured on. Per-step cost is flat in the bond dimension, against the 64-fold spread a χ^2 model predicts, so capacity is nearly free, chain length is the cost, and a resource estimate scales with the feature count rather than the ansatz. No QPU is required (Section 7).

Latency, against the budget the statement gives. Of the 300 ms authorisation ceiling about 130 ms is network, leaving roughly 170 ms issuer-side. Scoring one authorisation costs 0.1621 ms, 0.3110 ms at the tail; scoring one in-band authorisation against a fixed 64-row support set — one Gram row, not one kernel evaluation — would cost 92.062 ms, 568 times as much, its tail taking 66.8 % of that budget. Cost is linear in the support set, so the classical core is not the latency risk and the kernel is; the band bounds aggregate kernel compute, not the per-request tail, which a transaction landing in the band pays in full. Both bound model cost on this workstation at the profile `latency.csv` records for each — the scorer pinned to one thread, the kernel on unconstrained BLAS — so the ratio is not like for like.

Data, and why the PoC cannot reuse this file. The method needs settled labels and an entity key to cluster on; both are ordinary in an issuer’s data and only approximated here, and the benchmark’s competition rules permit non-commercial research use only. The protocol ports; the data does not.

Constraints we have measured, and would carry into Phase II. The population is conditional on an incumbent: this file contains only transactions an existing stack approved, because declined attempts cannot generate a chargeback. That is exactly the population a champion/challenger comparison runs on, so it is the right restriction — but no result extrapolates past it. The label is partly an entity flag, since a chargeback propagates to later transactions sharing an account, email or billing address, and Shapley attribution makes that concrete: 0.758 of the in-band model’s total contribution falls on `card1` alone, across 8 features. A governance review would read that as a model which mostly recognises cards, which is the strongest argument for running this on an issuer’s own data, where an entity key is a key and not a predictor. The certificate is sample-starved by design: conditioning on a narrow band is what makes the guarantee bind the component we care about and what limits how tight α can go. A larger calibration block or a wider band buys a tighter one; the method is unchanged.

What would change the quantum conclusion. An encoding demonstrably outside the RBF family, at a

bandwidth that does not concentrate, would pass the same screens unmodified; we did not find one. For the tensor network the requirement is a comparison powered for cross-family noise, about 7.77 times the in-band evaluation set. And the negative is about aggregate discrimination, not the operating point: H2 never ran, and a method that does not improve ranking can still improve a decision at a low false-positive threshold.

6 Validation plan

Most steps below were executed in this study on public data (Section 3); the power-sized evaluation block and the change-management cadence describe Phase II and were not. Some pre-registered items were *not* executed — two kernel hypotheses and the entanglement ablation, closed by the screens, and amendment A7’s drift measurements — and the appendix enumerates them, because a plan that cannot say which steps ran is not one.

Success is a certificate, not a score. A PoC succeeds if the band-conditional false-decline rate is certified at a level the business names, with family-wise error control, on a block never used to build it. Average precision and ROC AUC are reported alongside but decide nothing. Failure is equally well defined — the admissible set comes back empty, as it does here at the tightest band budget — and the acceptance test is the held-out read in Section 3.

Size the evaluation block to the power target first. A detectable-effect ceiling is fixed *before* the quantum-versus-classical comparison and the block sized to it — the one change that would make it informative here; Section 4 reports what happens when it is not.

Threshold change management. λ moves on every recalibration, so it needs an owner and a cadence rather than a rerun. Model risk approves each move quarterly, matching the rolling window the fraud-rate definition already uses, and out of cycle whenever realised in-band risk crosses the certified α . If a recalibration fails to certify, the rule falls back to the previously certified λ , and failing that to the incumbent scorer with the band closed — today’s behaviour, not an uncertified threshold. Phase II runs the challenger in shadow first, scoring every authorisation and acting on none.

What keeps the plan honest. Grids, splits and thresholds are hashed before any model is fitted, the hypotheses are frozen with dated amendments, and every retraction is recorded — 161 decision entries, including ones that removed results already written down.

7 Where the quantum method sits, and why there

Classical on all traffic, quantum on the band, conformal on the decision. A gradient-boosted scorer runs on every transaction at 0.8837–0.8884 ROC AUC over five seeds, because a hybrid whose classical half is weak proves nothing about its quantum half. The band is where a quantum model can go and the only place the arithmetic works: it holds thousands of rows rather than hundreds of thousands, and a kernel whose Gram matrix costs $O(n^2)$ circuit evaluations is affordable only on thousands. The certificate then wraps whatever scores the band, which is why the deliverable survived a negative quantum result — as it had to here.

The compute path. Both arms run on simulators, which the statement permits: no number here comes from a QPU. The fidelity kernel’s reference implementation is an exact statevector, with Braket’s `LocalSimulator` and Aer on CPU and GPU agreeing to within 2.887e-15 over 12 comparisons; the tensor network runs in PyTorch on one GPU. On the sample count the statement asks for: the screens ran on 300 stratified rows per configuration across 120 configurations, and the kernel arm was rejected before any downstream evaluation. The circuits are within near-term reach — at most 8 qubits, transpiled depth 35 and 28 two-qubit gates on a portable basis — so what stopped this arm was the screens, not the hardware.

8 Team

Amon Koike — Quantum Research Engineer, trading as Quantum Daemons, a sole proprietorship registered in Japan. B.S. Physics, Tokyo Denki University. Sole author of this submission. Portfolio: thedaemon-wizard.github.io/portfolio-quantum-research-engineer.

The statistical machinery is not new to the author. In QIntern 2026 Project 12 (QWorld), within Team A — which owns the paper’s conformal-coverage proposition — he wrote the split-conformal calibration of the false-alarm rate and the results freeze, whose manifests re-verify clean and whose suite was green at 135 on handoff. *Protocol-final and numbers-provisional in the team’s own words — the quantum side runs on a placeholder interface — so it establishes practice, not code: the implementation here was written from the published papers.*

Prior results. *Tensor networks:* Yale Peaked Hackathon 2026, 13th of 549 at 450 of 550 with team MerQury; the matrix-product-state attack is his own, merged upstream, exact to 50 qubits. *Quantum kernels:* QPoland 2025 runner-up with team The Cats Cradle. *Hardware:* VQC and QSVM on IBM `ibm_fez`, June 2026. *Upstream:* three pull requests merged into CUDA-Q and QuEST, two of them unitaryHACK 2026 bounties.